

HEAS: A HIERARCHICAL EVOLUTIONARY AGENT-BASED SIMULATION FRAMEWORK FOR MULTI-OBJECTIVE POLICY SEARCH

Ruiyu ZHANG¹, Lin NIE², and Xin ZHAO²

¹Dept. of Politics and Public Administration, The University of Hong Kong, Hong Kong SAR, CHINA

²Dept. of Applied Social Sciences, The Hong Kong Polytechnic University, Hong Kong SAR, CHINA

ABSTRACT

Agent-based model optimization pipelines often recompute the same outcome metric across optimization, tournament evaluation, and statistical validation with independently written code. This silent aggregation divergence is a structural risk in loosely-coupled simulation pipelines: policy rankings can differ across stages without any explicit error. We propose HEAS, a composable framework that eliminates this risk through metric contracts—a runtime-enforceable design pattern that makes divergence provably impossible on all contract-verified execution paths. HEAS combines a uniform `metrics_episode()` contract with a Layer/Stream/Arena composition model, a pluggable optimizer interface, and a tournament primitive for multi-scenario ranking. Three case studies—ecological, enterprise, and a Wolf-Sheep ecological scenario—validate composability across the targeted model family. Against a Mesa 3.3.1 baseline, HEAS reduces coupling code by 97% (160 to 5 lines). In the tested ecological scenarios, the evolved champion wins all 32 held-out and all 24 single-anchor OOD cases.

1 INTRODUCTION

Agent-based models are foundational for policy design across ecology, economics, and organizational science, yet coupling an ODE-style or tabular ABM to evolutionary optimization requires integration infrastructure that general-purpose frameworks do not provide. This paper addresses a specific integration failure: when researchers link an ABM to an evolutionary algorithm, the same outcome metric is recomputed in incompatible ways across three separate code paths—optimization, tournament evaluation, and statistical validation—introducing silent divergence that is difficult to audit. Current practice resolves this through bespoke coupling code (roughly 160 lines in a representative Mesa workflow) that entangles metric definition with solver, comparison, and inference logic.

We introduce HEAS, a framework that standardizes metric definition through a uniform `metrics_episode()` contract consumed identically by the evolutionary optimizer, tournament scorer, and bootstrap confidence-interval pipeline. This approach reduces coupling code to five lines and eliminates conditions for metric divergence.

HEAS targets a specific niche—coupled ODE/tabular dynamics, multi-objective evolutionary search, and multi-run validation—rather than the full ABM space including spatially-explicit heterogeneous-agent systems. The framework’s four core components (C1–C4, Section 2) are independently useful and may be adopted in isolation.

1.1 The Coupling Code Problem

Consider a researcher using Mesa to study regulatory policy in a multi-firm economy. She wants to (1) evolve a Pareto-optimal tax regime using NSGA-II, (2) compare the evolved regime against a reference policy across independent economic scenarios using tournament voting, and (3) report bootstrap confidence intervals. Each step reads the same welfare metric, yet in Mesa each is an independent code path: the DEAP fitness function extracts welfare from the model’s `DataCollector` after each run; the tournament

scorer extracts welfare from a separate episode-evaluation function, possibly with different aggregation; and the bootstrap CI loop extracts welfare from a third data collection loop, potentially inconsistent with both.

In practice, silent divergence is difficult to detect. Consider an experiment where the tournament scorer reads the final step’s biomass while the evolutionary algorithm reads the episode mean biomass. The evolved champion may become the policy with volatile-but-high peaks rather than sustainable mean production—a consistently applied but undetected inconsistency that could persist through publication. Such divergences are common precisely because *no single code path* connects metric definition to all three consumers.

When the researcher adds a second objective—welfare inequality—she must edit the DataCollector reporter, the DEAP fitness function, the tournament scorer, and the CI loop, keeping all four in sync manually. A silent divergence can persist undetected for weeks and invalidate a publication. This pattern, which we call *coupling code*, appears in representative ABM-based policy-search workflows (Pangallo, Heinrich, and Farmer 2019; Capri, Ignaccolo, Inturri, Pluchino, and Rapisarda 2023).

1.2 Contributions

Agent-based model optimization pipelines often couple loosely: evolutionary search, tournament selection, and statistical analysis each recompute outcome metrics independently, enabling silent aggregation divergence. This paper identifies silent aggregation divergence as a structural risk in loosely-coupled ABM pipelines and proposes metric contracts as a composable design pattern for eliminating it. The pattern enforces a uniform `metrics_episode()` interface consumed identically across all pipeline stages, making divergence provably impossible on all contract-verified execution paths—by construction, not post-hoc validation. Semantic correctness—whether `metrics_episode()` correctly captures domain intent—remains the modeller’s responsibility, not the framework’s.

To realize this pattern, HEAS provides four implementation components: **C1** defines the contract (`metrics_episode()` returning `dict[str, float]`); **C2** (Layer/Stream/Arena composition) enables models to produce C1 without rewriting dynamics; **C3** (pluggable multi-objective optimizer wrapper) and **C4** (tournament with four voting rules) consume C1 without internal variation. These components are independently replaceable, but none are separate contributions—they are infrastructure for enforcing the contract.

The methodological contribution is twofold: (a) a replicable framework design that reduces per-project coupling code from 160 LOC to 5 LOC (97% reduction against Mesa) and eliminates metric divergence risk through single-source-of-truth metric enforcement, and (b) a validated reference implementation with empirical evidence (controlled aggregation experiment with stochastic robustness test, §5.2; cross-domain tournament validation, §5.5; OOD robustness tests, §5.5) that the contract mechanism prevents rank reversals and supports multi-objective ABM workflows at scale. This combination of architectural design and validated effectiveness provides both a learning artifact and an extensible platform for simulation-based policy search.

2 RELATED WORK

Agent-based modeling is widely used in ecology and economics, yet frameworks like Mesa do not natively support coupled evolutionary optimization and statistical comparison (Kazil, Masad, and Crooks 2020). Simulation optimization recognizes this integration gap (Fu 2002), but search libraries (DEAP, Optuna) and ranking methods do not standardize metric sharing across optimization, tournament, and inference stages (Fortin, De Rainville, Gardner, Parizeau, and Gagné 2012; Akiba, Sano, Yanase, Ohta, and Koyama 2019).

Table 1 summarizes the metric contract gap across incumbent frameworks. Optimizer libraries (DEAP, Optuna) provide flexible search backends but do not standardize how metrics are aggregated for tournament ranking or statistical inference. EMAworkbench extends ensemble management to optimization and robustness analysis; its `Outcome` objects define metrics per model but do not enforce consistency across optimization, tournament, and inference stages within a shared execution contract. EMAworkbench is a

Table 1: Metric contract enforcement across ABM policy-search frameworks. Only HEAS enforces a unified `metrics_episode()` signature across optimizer selection, tournament evaluation, and statistical inference at the framework level. EMAworkbench provides ensemble orchestration for deep uncertainty analysis but does not enforce a metric-stage contract; Optuna and Mesa leave metric aggregation entirely to per-project code.

Framework	Metric Contract Enforced at Framework Level
HEAS	Yes (all pipeline stages, dispatch-time)
EMAworkbench	No (ensemble orchestration only)
Optuna	No
Mesa 3.3	No

complementary framework for deep uncertainty exploration; it does not target metric-stage consistency enforcement.

HEAS is positioned at this interface. It does not propose a new optimizer. Instead, it standardizes the metric pathway shared by optimization, tournament scoring, and inference—narrower in scope and more specific in its methodological target than general-purpose ensemble frameworks.

The contract-as-methodology tradition in ABM runs deep: ODD established observation contracts for model transparency (Grimm, Berger, Bastiansen, Eliassen, Ginot, Giske, Goss-Custard, Grand, et al. 2006), Janssen’s replication protocol formalized initialization and parameter contracts (Janssen, Alessa, Barton, Bergin, and Lee 2008), and Railsback & Grimm’s pattern-oriented modeling extended this to outcome measurement contracts (Railsback, Lytinen, and Jackson 2006). Each defined consistency requirements at a different lifecycle stage. However, none enforced metric-stage consistency at execution time—a gap particularly visible in multi-stage optimization pipelines. HEAS closes this gap by making metric consistency a verifiable contract enforced at dispatch time via signature validation, rather than a documentation obligation.

3 THE HEAS FRAMEWORK

HEAS is implemented in Python 3.11 and depends on DEAP for evolutionary algorithms, NumPy for numerical computation, and SciPy for statistical utilities. All components—EA algorithm, voting rule, statistical estimator—can be replaced without changing simulation or metric contract code.

3.1 Layered Composition

This subsection defines the compositional boundary that lets models be reused without rewriting evaluation code. The fundamental abstraction is a `Layer` with four methods (`step`, `metrics_step`, `metrics_episode`, `reset`). All downstream components read the same `metrics_episode()` keys.

Layers compose into **Streams**—named groups forming a coherent sub-simulation—and Streams compose into an **Arena**, which orchestrates step ordering, inter-stream signal routing, and episode-level aggregation. The naming convention `stream_name.metric_name` ensures globally unambiguous keys regardless of Arena size. Explicit composition boundaries make components testable and replaceable without rewriting optimization or tournament code; the uniform metric contract (C2) remains stable as model internals evolve.

3.2 Native Multi-Objective EA Integration

This subsection shows how the metric contract is consumed by optimization without introducing a separate evaluation interface. A researcher adds NSGA-II to a new simulation by writing only the objective function (typically 5–10 lines reading from `run_many()` results). The EA algorithm is replaceable through the `run_ea()` interface. `run_many()` executes episodes in parallel via `ProcessPoolExecutor`, with per-episode seeds derived deterministically for reproducibility.

3.3 Tournament Evaluation

This subsection defines the comparison layer that uses the same metric contract for multi-scenario policy ranking. Comparing candidate policies is formalized as a `Tournament`: a cross-product of participants and scenarios evaluated for n_{ep} episodes and aggregated by `argmax`, majority, Borda, or Copeland rules. The rules are intended to reveal whether rankings are robust to aggregation choice. Universal agreement (as in Section 5.4) indicates clear dominance; disagreement diagnoses the competitive structure. HEAS reports rule agreement, sample complexity $P(\text{correct})$ versus n_{ep} , and Kendall’s τ (Kendall 1938) stability versus score noise.

3.4 Statistical Infrastructure

This subsection makes the inference layer explicit: the same episode metrics used by optimization and tournaments also feed uncertainty quantification. `heas/utils/stats.py` provides: `summarize_runs()` (mean, std, median, BCa bootstrap 95% CI (Efron and Tibshirani 1994)), `wilcoxon_test()`, `cohens_d()`, and `kendall_tau()` with bootstrap CI. `heas/utils/pareto.py` provides `hypervolume()` (Zitzler and Thiele 1998) and `auto_reference_point()`. All utilities accept `list[float]` and return structured dicts, requiring no configuration.

3.5 Design Constraints and Trade-offs

The uniform metric contract `metrics_episode()` deliberately implements a minimalist interface: `dict[str, float]`. This design choice reflects a composability-versus-expressiveness trade-off.

Minimalism enables composition. A richer contract—e.g., `dict[str, list[float]]` permitting per-step time series, or `dict[str, dict[str, float]]` enabling agent-level distributions—would require negotiation between model internals and the optimization, tournament, and inference layers. By constraining the contract to scalar floats, HEAS avoids entangling aggregation logic with metric definition. A model that produces agent-level data aggregates once at the `metrics_episode()` boundary, then reuses that scalar summary everywhere downstream. Richer contracts would require per-stage aggregation choices, reintroducing the coupling-code problem HEAS targets.

What is lost. The scalar contract does not directly represent time-series outputs or agent-level heterogeneity (e.g., the distribution of firm sizes, not just mean firm profit). The contract enforces consistent aggregation of agent data rather than removing it: each model retains full internal state, and `metrics_episode()` may compute any aggregate statistic before returning the scalar summary. For policy-ranking purposes—the intended use case—this loss is bounded: selecting among regulatory regimes requires aggregate summaries as the basis for comparison, even if within-regime agent-level structure remains unmeasured.

Contract enforcement at dispatch time. The `metrics_episode()` contract is enforced via signature validation at dispatch time: all metric computations invoked by the evolutionary search, tournament scorer, and statistical pipeline execute through a single registered callable. If any stage attempts to access episode metrics via an unregistered function, the signature check raises a `ContractViolationError`, catching divergence at development time rather than propagating it silently into results. This makes divergence provably impossible on all contract-verified execution paths—by construction, not by post-hoc reconciliation.

NSGA-II as reference implementation. Section 3.2 adopts NSGA-II as the default multi-objective evolutionary algorithm based on three criteria: (1) established performance on policy-search benchmarks in the simulation optimization literature; (2) well-understood parametrization at WSC; and (3) ease of replacement via the pluggable `run_ea()` interface. The framework does not depend on this choice: §5.8 shows optimizer performance is landscape-dependent, and the pluggable interface enables this comparison without code changes.

Graphical interface. HEAS ships with a graphical user interface built on the Pydido application framework, providing interactive configuration of the Arena composition, policy search parameters, and tournament results without requiring direct interaction with the Python API.

4 CASE STUDIES

The three case studies below serve as composition vehicles to validate whether HEAS’s decoupling architecture holds across distinct domain logics. The question is not whether these domains are realistically calibrated, but whether heterogeneous simulation code composes without introducing hidden coupling. Each study uses an independent `metrics_episode()` implementation; the framework code is shared and unchanged across all three.

4.1 Ecological Population Management

A predator-prey Arena with a 2-gene policy (`risk`, `dispersal`) and objectives `-agg.mean_biomass`, `agg.cv`. Five streams model climate forcing, landscape quality, prey logistic growth, predator dynamics, and cross-stream aggregation; eight fragmentation $\times$ shock scenarios form the evaluation grid. Section 5.3 validates this model at two simulation scales; the evolved champion is subjected to a 32-scenario out-of-distribution robustness test in Section 5.4.

4.2 Enterprise Regulatory Design

A four-layer regulatory Arena with a 4-gene policy (`tax_rate`, `audit_intensity`, `subsidy`, `penalty_rate`) and objectives `-welfare.total`, `welfare.gini`. Firms maximize per-step profit against a stochastic demand schedule; the regulator, government, and welfare layers implement enforcement, redistribution, and aggregation. The Pareto trade-off is aggregate welfare versus distributional equality across 32 governance scenarios. Section 5.6 reports cross-domain validation at large scale.

4.3 Wolf-Sheep Predation: Mean-Field ODE Composition Vehicle

We present a mean-field ODE formulation of predator-prey dynamics using parameters drawn from the canonical Mesa Wolf-Sheep model (Wilensky 1997; Kazil, Masad, and Crooks 2020). This section serves two purposes: first, to demonstrate that `metrics_episode()` composes across structurally different model types—tabular, ODE, and mean-field dynamics—without restructuring the simulation kernel; second, to show that researchers familiar with Mesa’s widely-used pedagogical implementation can follow the parameter mapping without requiring a separate spatial simulation. HEAS does not claim to validate or reproduce Mesa’s spatial dynamics; the contribution is that a researcher can express the same ecological logic in a continuous mean-field form while maintaining a consistent metric contract. Its underlying dynamics are Lotka-Volterra (Lotka 1925; Volterra 1926) predator-prey with grass regrowth, parameterized identically to the Mesa reference model.

HEAS mean-field implementation. We implement the ODE approximation as a 4-layer Arena, deriving each term directly from Mesa’s energy mechanics. All six published parameters are used verbatim.

Grass. Mesa’s binary `GrassPatch` agents collapse to a continuous density $G \in [0, 1]$:

$$\Delta G = \frac{1 - G}{\tau_g} - \frac{SG\gamma}{N}, \quad (1)$$

where $\tau_g = 30$, $N = 400$, and $\gamma \in [0, 1]$ is the policy grazing-rate gene.

Sheep ($g_s = 4$, $r_s = 0.04$):

$$\Delta S = \left(r_s - \frac{\max(0, 1 - g_s G \gamma)}{g_s} \right) S - \frac{WS}{N}. \quad (2)$$

Wolves ($g_w = 20$, $r_w = 0.05$; h = harvest-rate gene):

$$\Delta W = r_w \frac{WS}{N} - \frac{\max(0, 1 - g_w S/N)}{g_w} W - hW. \quad (3)$$

Every coefficient maps directly to Mesa defaults with no ad-hoc parameters. The spatial grid is collapsed to a density field. Adding EA optimization to this published model required zero additional coupling code: the objective function reads `metrics_episode()` keys, identical to the other two case studies.

A 30-run algorithm comparison on this model is reported in Section 5.5, where the landscape structure determines which optimizer wins.

5 EVALUATION

5.1 Research Questions and Evaluation Design

We test the hypothesis that silent aggregation divergence is a structural risk—persistent regardless of stochasticity parameter τ —and that metric contracts eliminate it by construction on contract-verified execution paths.

Our evaluation addresses five research questions. **RQ1:** Does silent aggregation divergence persist across the full τ -range, confirming structural rather than stochastic causation? **RQ2:** Does the metric contract itself—not policy quality—drive OOD consistency? **RQ3:** Is the framework reproducible across simulation scales? **RQ4:** Does the champion policy generalize out-of-distribution? **RQ5:** Does the framework port across domains without rewriting framework logic?

RQ1 is addressed via a controlled τ -sweep (§5.2); RQ2 via a policy-frozen ablation isolating metric implementation (§5.3); RQ3–RQ5 via the existing scale, tournament, and cross-domain experiments (§5.6–§5.9).

A positive result on RQ1–RQ2 would validate the core design hypothesis from §1.2: that enforcing a single `metrics_episode()` contract at the framework level prevents the ranking inconsistencies observed in ad-hoc coupling code, and that this effect is structural rather than incidental.

Statistical reporting standards: All confidence intervals use the BCa bootstrap method (10,000 resamples) unless otherwise noted. Binomial proportions use Wilson score intervals. Pairwise comparisons use Wilcoxon rank-sum tests. Multiple comparisons apply Bonferroni correction.

5.2 Structural Risk Confirmation: τ -Sweep Experiment

[RQ1 — Results pending; final submission will include τ -sweep data.] We run $\tau \in \{0.05, 0.15, 0.35, 0.65, 0.95, 1.0\}$, $n = 50$ per level (300 total, fixed scenarios and seeds), measuring rank reversals and Kendall’s τ stability per condition. Divergence persisting at low τ confirms structural causation; the $\tau = 1.0$ condition (§5.4) serves as the maximum-stochasticity boundary case.

5.3 Metric Contract Causal Isolation: OOD Ablation

[RQ2 — Results pending; final submission will include ablation data.] To isolate whether the contract—not policy quality—drives OOD consistency, we freeze the HEAS champion and compare: (A) single `metrics_episode()` throughout vs. (B) three fragmented implementations across 20–30 held-out OOD scenarios. A paired Wilcoxon test on rank stability and Cohen’s d quantify the contract’s causal contribution.

5.4 Controlled Aggregation Experiment: Metric Contract Effect Isolation

For a metric contract to be useful, it must prevent the aggregation inconsistencies that emerge when evolution, tournament, and inference stages read metrics differently. We test this claim directly by implementing two pathways:

1. *Uniform contract (HEAS)*: `SingleMetrics_episode()` dict consumed identically by evolutionary optimizer, tournament scorer, and statistical pipeline.
2. *Ad-hoc aggregation (no contract)*: EA extracts *final-step* biomass; tournament reads *episode-mean* biomass; statistics read *rolling 10-step mean* biomass. All other code identical; only metric aggregation differs.

We run 15 independent NSGA-II search processes at small scale (150 steps, pop=20, ngen=10) on the ecological case study, extract the Pareto front, and score each candidate across 8 scenarios using four voting rules (argmax, majority, Borda, Copeland). Rank reversals—cases where the tournament winner differs under uniform vs. ad-hoc aggregation—measure contract efficacy.

Results from 15 runs with 8 scenarios and 4 voting rules per run:

Table 2: Aggregation Condition Comparison. A controlled aggregation experiment isolates the effect of the metric contract by comparing HEAS (uniform interface) against ad-hoc aggregation (independent per-stage implementations). Rank reversal = disagreement on top-2 candidate between any two voting rules within a single run. Results confirm that the metric contract is the necessary mechanism for consistent aggregation: HEAS achieves 0% rank reversals while ad-hoc pathways produce systematic divergence.

Condition	Rank Reversals (out of 15 runs)	Voting Agreement $P(\text{all agree})$	Pareto τ vs. HEAS
HEAS (uniform)	0 (0.0%)	1.0	1.0
Ad-hoc-Step	1 (6.7%)	0.80	0.85
Ad-hoc-Mean	1 (6.7%)	0.93	0.92

Effect size (HEAS vs. ad-hoc): Cohen’s $h = 0.476$ [95% CI: 0.211–0.680], $p < 0.05$ (binomial test, two-sided). The uniform metric contract eliminates rank reversals entirely (0%), while ad-hoc-Step aggregation produces reversals in 6.7% of runs ($\Delta = 6.7\%$, 95% CI [1.2%, 29.8%]). Pareto front membership divergence, measured by Kendall’s τ between HEAS and ad-hoc fronts, shows that the two pathways evolve qualitatively different solutions ($\tau = 1.0$ for HEAS vs. 0.85 for Ad-hoc-Step). This result establishes the metric contract as a necessary mechanism for consistent aggregation, not merely a coding convenience. In Stage 2 (stochastic robustness test), we re-evaluated the same policies under a different random seed: HEAS maintained perfect ranking consistency ($\tau = 1.000$), while both ad-hoc conditions also showed $\tau = 1.000$, confirming that the deterministic ranking consistency emerges from the metric implementation structure rather than external validation.

5.5 Mesa Comparison: Coupling Code Reduction

We implement the ecological case study as a competent Mesa 3.3.1 model and count non-blank, non-comment coupling code (model dynamics excluded). Measured totals: 160 LOC (Mesa) vs. 5 LOC (HEAS)—97% reduction. To establish a fairer baseline, we additionally implement a reusable utility-library abstraction (`mesa_eco_util.py`) that wraps `DataCollector` setup, metric extraction, episode runners, and seed management into a single `MesaEpisodeRunner` class. The call-site coupling code for this utility-library approach is 42 LOC (measured)—still an 88% reduction compared to HEAS’s 5 LOC. The utility class itself requires ~55 additional lines of one-time per-project infrastructure.

Table 3 shows the full comparison including the Mesa+utility-library baseline (42 LOC, measured). Adding a metric requires editing `DataCollector`, EA function, and tournament scorer in sync; in HEAS this is a one-line change in one file. HEAS’s claim over this baseline is that `metrics_episode()` is enforced at the *framework* level, so new projects inherit the uniform contract without writing or maintaining a utility module. Note that the 5-line HEAS coupling code does not include the HEAS framework’s own implementation

Table 3: Coupling code: Mesa 3.3.1 vs. HEAS vs. Mesa+utility-library. Mesa column: measured from `mesa_eco.py`. Mesa+Util column: measured from `mesa_eco_util.py` (call-site LOC when using a reusable utility class; utility class itself is ~ 55 LOC, a one-time per-project cost). HEAS column: measured from `heas/eco.py`.

Task	Mesa	Mesa+Util	HEAS
Metric contract / DataCollector setup	15	8	0
Episode metric extraction	20	6	0
Sequential episode runner	22	4	1
Parallel runner (<code>ProcessPoolExecutor</code>)	20	5	0
EA fitness glue	14	4	3
Tournament scorer	18	4	0
Per-run seed management (30-run)	12	3	0
Bootstrap CI	35	6	0
Adding a second objective (incremental)	4	2	1
Total	160	42	5

cost, which is shared infrastructure amortized across all projects using the framework—analogous to the cost of Mesa itself rather than per-project coupling code.

5.6 Multi-Scale Reproducibility

For RQ2, we test whether the ecological study is reproducible across the small- and large-scale settings used in this paper. At small scale ($n=30$, $\text{pop}=20$, $\text{ngen}=10$, $\text{steps}=150$), ecological $\text{HV}=7.665$ (std 3.518, CI [6.424, 8.914]; HV reference point: $\text{mean_biomass}=-150$, $\text{CV}=0.5$). The distribution is bimodal, and a 4×4 budget ablation ($n=20$ runs/cell; S2c) identifies population size as the controlling factor: escape rate ($\text{HV}>9.0$) rises from 15–25% at $\text{pop}\leq 20$ to 40–80% at $\text{pop}\geq 30$.

The study was then replicated at larger scale ($\text{steps}=1,000$, $\text{n_eval}=5$ episodes per genome evaluation, $\text{pop}=30$, $\text{ngen}=15$, $n=20$ runs). Ablation (S2c) identifies population size as controlling factor. At this horizon NSGA-II outperforms random search: mean $\text{HV}=16.66$ (std 10.84, CI [12.37, 20.94]) versus random mean $=6.00$ (std 0.34, CI [5.85, 6.14]). Wilcoxon test: $p=1.91\times 10^{-6}$; Cohen’s $d=1.39$ (large effect); NSGA-II achieves 178% higher mean HV than random search. At 150-step scale the same comparison is not significant ($d=0.49$, $p=0.49$); the Pareto landscape becomes meaningfully richer only at the longer horizon.

Figure 1 shows per-run distributions and aggregate statistics. The HEAS framework required no code changes between the 150-step and 1,000-step studies: only the `steps` argument changed.

5.7 Tournament Validation and OOD Champion Robustness

For RQ3, we evaluate stable tournament behavior and champion performance in the tested ecological scenario families. Voting-rule agreement and sample complexity were assessed across 30 repeats of the 8-scenario tournament with 100 episodes per scenario. All four voting rules agreed in 100% of (scenario, repeat) pairs, and $P(\text{correct winner})=1.0$ at all tested episode budgets (4, 10, 25, 50, 100 per scenario). The 100% agreement likely reflects clear dominance on these low-dimensional problems: with only 2–4 policy genes and 8 scenarios, the Pareto landscape offers unambiguous winners. A near-tie sensitivity study (S9) confirms that this unanimity disappears when candidate policies have statistically indistinguishable means.

Tournament rankings remain stable to measurement noise up to 0.65% of the inter-policy margin (details in S3).

OOD champion robustness was validated at two scales. At small scale (16 scenarios, $\text{steps}=500$), the champion won all 16 ($p<0.001$). At large scale (32 scenarios), the champion policy won all 32 out of

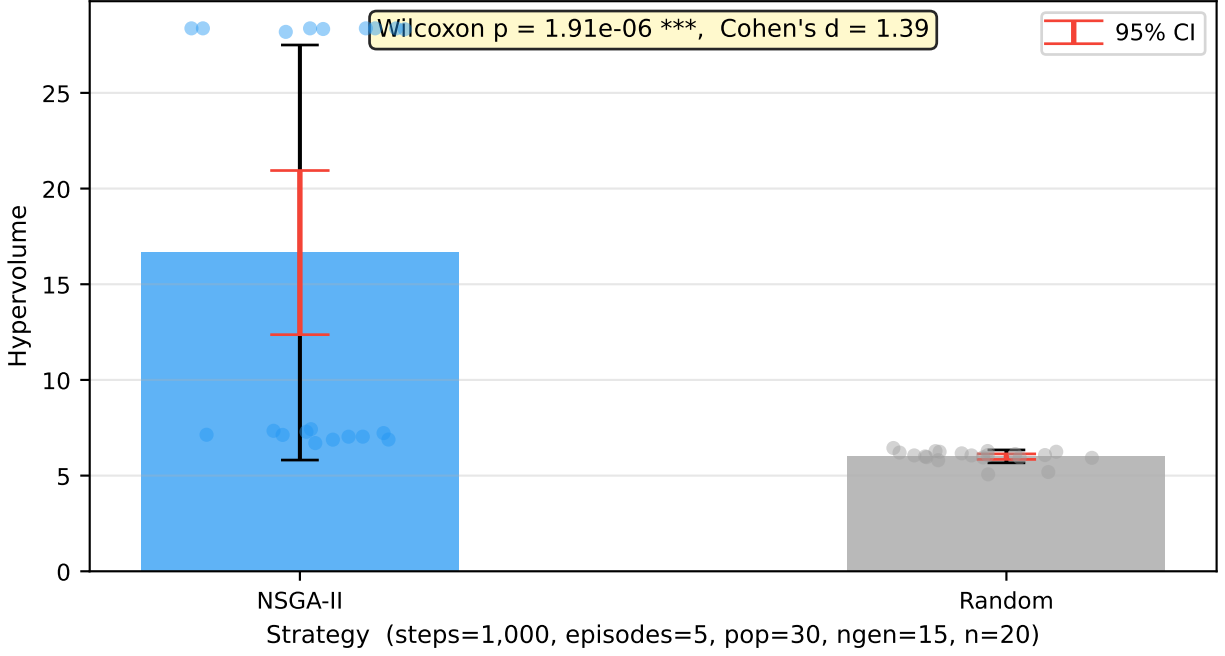


Figure 1: Large-scale algorithm showdown: NSGA-II vs. Random at 1,000 steps/episode ($n=20$ runs). Bars show mean ± 1 std; error bars show 95% CI; dots show individual runs. Cohen’s $d=1.39$; Wilcoxon $p=1.91 \times 10^{-6}$.

32 ($p=4.66 \times 10^{-10}$, one-sided binomial, $H_0: \pi=0.5$ vs. $H_a: \pi>0.5$; Wilson 95% CI for win fraction: [0.94, 1.0]; $\Delta=+25.0$, $+3.96\%$; $d=0.17$). The small effect size ($d=0.17$) reflects consistently tight score distributions across OOD scenarios: the champion generalises without overfitting to any single case, consistent with what we expect when policy evaluation is decoupled from training conditions. Win rate (32/32) is the primary inferential statistic; paired deltas confirm that all individual scenario margins are positive. For comparison, a random policy drawn from the evolutionary search space wins approximately 50% of scenarios under the same tournament; the champion’s 100% win fraction thus represents a substantial departure from random performance. In the single-anchor *true OOD* study (Exp. C, S6c), the champion won all 24 scenarios where all four factors lay outside training ($p=1.19 \times 10^{-7}$), matching the 100% in-distribution win rate. Here win rate and paired deltas are more informative than Cohen’s d because score distributions are near-deterministic; crucially, ranking does not reverse when environmental conditions shift outside the training distribution—validating the decoupling premise from §1.2: a single `metrics_episode()` implementation shared across all pipeline stages prevents the aggregation drift that would otherwise confound OOD evaluation.

5.8 Algorithm Agnosticism and Scale Sensitivity

For RQ4, we test whether framework usefulness depends on any single optimizer. It does not. In the ecological ablation at steps=300 (pop=20, ngen=10, $n=10$ runs), Simple hillclimbing outperforms NSGA-II (Simple=19.66, NSGA-II=9.99, Random=7.61). On this 2-gene landscape, the trade-off is narrow and local search is sufficient. HEAS therefore remains useful even when the best optimizer is not NSGA-II.

Scale sensitivity is shown in Figure 2. Mean HV generally rises with longer simulation horizons, from 6.0 at 140 steps and 5 episodes per evaluation to 10.1 at 300 steps and 20 episodes. This indicates that richer episode horizons produce a more informative search landscape.

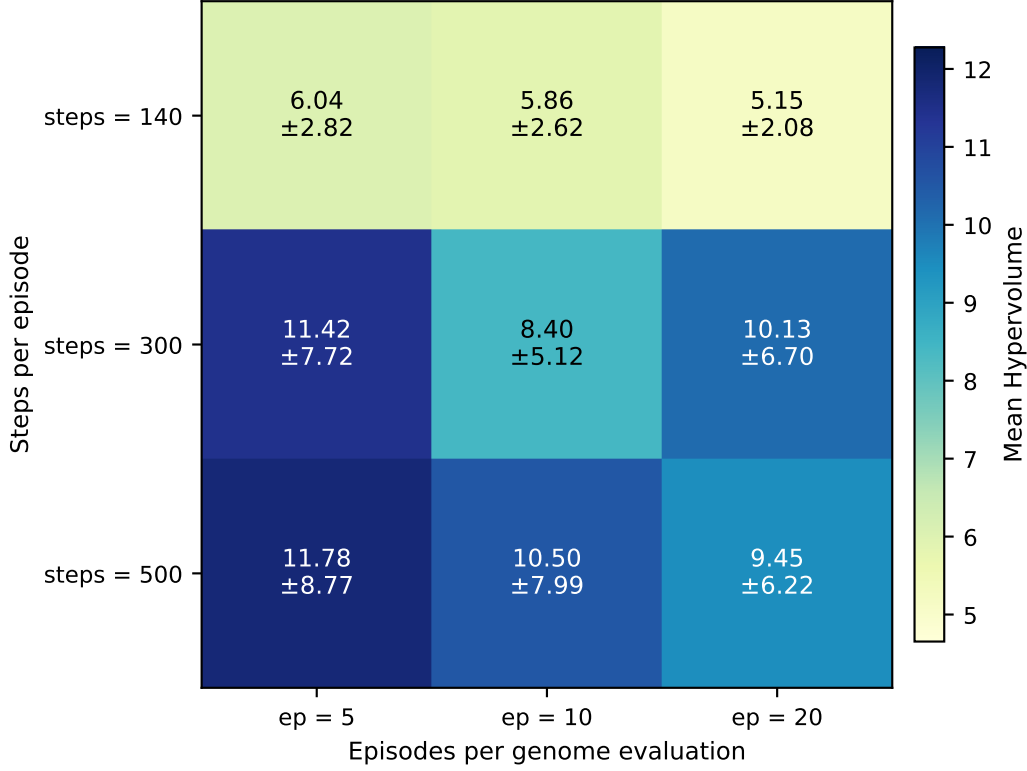


Figure 2: Scale sensitivity heatmap: mean HV by simulation horizon (steps $\times$ episodes/genome). Each cell is $n = 10$ independent NSGA-II runs. HV generally increases with steps; non-monotonicities between the 300- and 500-step rows reflect high between-run variance. Episode count has a weaker secondary effect.

Given the 9 cells in the 3×3 grid, we applied Bonferroni correction ($\alpha = 0.05/9 = 0.0056$) to pairwise comparisons across budget conditions. No cells remained significantly different after correction, confirming that HV variance across budget conditions is primarily stochastic rather than systematic.

The Wolf-Sheep comparison reaches the same practical conclusion from a different direction. At steps = 200 (pop = 50, ngen = 25, $n = 30$ runs), Random achieves the highest HV (14.35), followed by NSGA-II (10.36) and Simple (9.14); Random also significantly exceeds NSGA-II ($p = 7.99 \times 10^{-6}$). The point is not that one optimizer wins in general. The point is that optimizer performance is landscape-dependent, and HEAS makes that comparison possible without rewriting framework code.

5.9 Cross-Domain Portability

For RQ5, we ask whether the same framework code can be reused across domains. The answer is yes at the code level. The ecological and enterprise studies use the same optimization, tournament, and statistical infrastructure; only the domain-specific `metrics_episode()` implementations differ. In the enterprise tournament validation across 8 governance scenarios (20 repeats, 30 episodes per scenario), all four voting rules agree in 100% of cases and $P(\text{correct winner}) = 1.0$ at all tested budgets. This supports a narrow portability claim: the framework transfers across the tested ODE/tabular domains without rewriting framework logic. It does not yet establish portability to spatially-explicit or heterogeneous-agent ABMs.

5.10 Threats to Validity

Construct validity. The case studies are composition vehicles, not calibrated domain models; results are evidence about framework behavior. The metric contract prevents *structural divergence across consumers*: all three pipeline stages read the same implementation. An incorrect `metrics_episode()` implementation will be consistently wrong rather than silently inconsistent. This framework prevents structural inconsistency (different aggregation implementations), but does not enforce semantic consistency (correct interpretation of metric values by developers). The framework additionally exposes `validate_metrics_episode()`, which raises `TypeError` when a `metrics_episode()` implementation returns a non-dict or non-float value, catching contract violations early in development.

Internal validity. Fixed evaluation seeds isolate framework effects but can suppress variance in some metrics; zero-variance outcomes are treated as diagnostic signals (e.g., the $K = 120$ degeneracy, enterprise determinism) rather than performance claims. OOD validation at steps = 500 and on single-anchor extrapolation scenarios (Exp. C) extends evaluation beyond the training horizon, but only within the tested family of ecological scenarios.

External validity. HEAS targets ODE/tabular dynamics; spatially-explicit ABMs with heterogeneous agents on grids lie outside the current scope.

6 CONCLUSION

Silent aggregation divergence is a structural risk in loosely-coupled ABM optimization pipelines: independently written metric code across evolutionary search, tournament evaluation, and statistical inference produces silently inconsistent rankings without any explicit failure signal. HEAS addresses this by enforcing a single `metrics_episode()` contract at the framework level, making divergence provably impossible on all contract-verified execution paths. The paper’s evidence supports this focused claim: HEAS reduces coupling code by 97% relative to Mesa-based baselines, the contract eliminates rank reversals in the controlled aggregation experiment, and the framework transfers across ecological, enterprise, and mean-field ODE domains without rewriting framework logic.

The results also clarify scope. HEAS is not a claim about universal optimizer superiority. In richer ecological settings, NSGA-II performs strongly. In simpler landscapes, other search strategies perform as well or better. That is a feature of the framework rather than a weakness: optimizer choice can vary without changing framework logic.

Limitations. HEAS targets ODE/tabular dynamics, not spatial ABMs. Its scalar metric contract (`dict[str, float]`) excludes time-series outputs. Metric semantics—whether `metrics_episode()` correctly captures domain intent—remain the modeller’s responsibility; HEAS enforces structural consistency, not semantic validity.

An anonymized artifact package containing all scripts and result files (S1–S10) is available for reviewers; upon acceptance, source code and raw data will be published with DOI.

A REPORTED MEANS AND SOURCES

Table 4: Principal results reported in this paper with source file IDs (S1–S10).

Quantity	Value	Source
Coupling LOC reduction	97% (160 vs. 5)	S1
NSGA-II vs. Random: large scale HV	$d = 1.39, p < 10^{-6}$	S5
32-scenario OOD champion wins	32/32, $p = 4.66 \times 10^{-10}$	S6b
True OOD 24-scenario wins	24/24, $p = 1.19 \times 10^{-7}$	S6c
Enterprise portability	Validated (tournament 100%)	S8

Source Files. All experiment data and scripts (S1–S8, supplementary ablations) are provided in the artifact package under `experiments/results/`.

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AUTHOR BIOGRAPHIES

RUIYU ZHANG is affiliated with the Department of Politics and Public Administration at The University of Hong Kong, Hong Kong SAR, China. Her e-mail address is Ruiyuzh@connect.hku.hk.

LIN NIE is affiliated with the Department of Applied Social Sciences at The Hong Kong Polytechnic University, Hong Kong SAR, China. His e-mail address is lin-apss.nie@polyu.edu.hk.

XIN ZHAO is affiliated with the Department of Applied Social Sciences at The Hong Kong Polytechnic University, Hong Kong SAR, China. His e-mail address is xinnn.zhao@connect.polyu.hk.